Monte Carlo Methods for Hypothesis Tests: Recap

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Module 1: Integrate — Session 4

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Hypothesis Testing

Goal: use a dataset to decide between two options possible states of the world.

- H_0 (null) and H_1 (alternative) define two distributions (or classes of distributions)
- A **test statistic** $T = T(X_1, \ldots, X_n)$: one number summarizing the evidence against the null.
- A **rejection region** $\mathcal{R}$: the values of T that make us reject.

The decision is a Bernoulli random variable, so everything we want to know about the test is a **probability**.

Two ways to be wrong

	H_0 true	H_1 true
reject	**Type I error** (α)	correct ($1-\beta$, the **power**)
do not reject	correct ($1-\alpha$)	**Type II error** (β)

- You **set** α. Fix the size you will tolerate, then take all the power you can get.
- "Do not reject" is **not** "accept H_0" — a test with no power fails to reject anything.

MC estimates of size and power

$$\alpha = \mathsf{E}\left(\mathbf{1}(T \in \mathcal{R}) \mid H_0\right), \qquad \text{power}(\theta_1) = \mathsf{E}\left(\mathbf{1}(T \in \mathcal{R}) \mid \theta_1\right)$$

```
null <- replicate(k, T(rmodel(n, theta_0)))
alt <- replicate(k, T(rmodel(n, theta_1)))
# plot to determine rejection region,
# write function `is_in_R()` returns TRUE/FALSE
mean(is_in_R(null)) # size
mean(is_in_R(alt))  # the power
```

p-values

- One sided:

$$\hat{p} = \frac{1 + \#\{T_b \geq t_{\text{obs}}\}}{B + 1} \qquad \texttt{mean(c(t_obs, null_dist) >= t_obs)}$$

- Two sided:

$$\hat{p} = 2 \min \left\{ P(T \leq t_{\text{obs}}),\ P(T \geq t_{\text{obs}}) \right\}, \text{ capped at 1}$$

- More generally, order rejection regions largest to smallest, find α of first rejection region to include observed value.

Inverting a test gives an interval

Hold the data fixed and ask which hypotheses not rejected:

$$\mathrm{CI}_{1-\alpha} = \{\theta_0 : \text{the level-}\alpha \text{ test of } H_0 : \theta = \theta_0 \text{ does not reject}\}$$

- One simulated null distribution **per candidate** θ_0 on a grid; keep the ones you cannot reject.
- The proof is just the size of the test: if θ is the truth, the test rejects it with probability α, so the set contains it with probability $1 - \alpha$.